

Work Experience

Nov. 2023 – Current	Paris Northern Computer Science Lab (LIPN – UMR 7030) AI PhD Researcher • Scientific Mentorship: Master's 2 intern (EID2 track, Sorbonne Paris Nord University). In collaboration with Basarab MATEI.	Paris France
Nov. 2026 – Current	Conservatoire Nationale des Arts et Métiers (CNAM) Lecturer (ATER) • Virtualized and Cloud Systems — LP • System and Network Administration — DEUST • Networks — DSP • Office Software — CPDA	Paris France
Nov. 2023 – Oct. 2026	Sorbonne Paris Nord University (USPN) & Sorbonne University (SU) Part-time Teacher • Data Structure — L2 Computer Science — SU • Data Visualization — L3 Computer Science — USPN • Algorithmics — L3 Computer Science — SU • IMA Project mentorship — M1 Computer Science — SU	Paris France
Apr. 2023 – Sept. 2023	The Engineering science, Computer science and Imaging Laboratory (ICube – UMR 7357) AI intern Researcher • Supervisor: Thomas LAMPERT • Project: Representation Learning for Geographic Spatio-Temporal Generalization ◦ Propose a new Homogeneous Time Series Domain Adaptation (Domain Invariant Representation) architecture based on combining <i>Shapelets</i> and <i>Adversarial Discriminative Domain Adaptation (ADDA)</i> techniques. ◦ Analyze and identify success factors of Domain Adaptation techniques on <i>Satellite Images Time Series</i> data.	Strasbourg France
May 2022 – July 2022	Paris Northern Computer Science Lab (LIPN – UMR 7030) Research Engineer • Supervisor: Guénaél CABANES • Implementation of a benchmark for unsupervised data flow clustering algorithms. • Development of testing protocols to compare state-of-the-art approaches with algorithms developed by the laboratory team. • Outcome: This work contributed to the article published in the journal <i>“Pattern Recognition”</i> (2025).	Paris France
May 2021 – July 2021	Computer Science Faculty of Monastir – University of Monastir Computer Science Intern (Licence) • Supervisor: Aymen TRIGUI • Transfer learning applied to a convolutional neural network (CNN) to adapt it for mask-wearing detection. • Development and deployment of an access control system on a Raspberry Pi using the adapted CNN model.	Monastir Tunisia

Education

Nov. 2023 – Current	Sorbonne Paris Nord University PhD in Computer Science • Supervisor: Basarab MATEI • Title: Anatomical structures 3D segmentation and reconstruction using Self-Supervised Deep Learning	Paris France
Sept. 2021 – Sept. 2023	Sorbonne Paris Nord University Master's degree in computer science • Data Mining and Decision-Making (EID2) curriculum • PNGS-M&CS Graduate School scholarship • With Honors – Ranked 1st	Paris France
Sept. 2018 – Sept. 2021	Université de Monastir Licence en Informatique • With Honors – Ranked 1st	Monastir Tunisia

Publications		
✓	International peer-reviewed journals	
Jan. 2025	X. Qian, G. Cabanes, P. Rastin, M. A. Guidani, G. Marrakchi, M. Clausel, N. Grozavu. "Incremental clustering based on Wasserstein distance between histogram models". Pattern Recognition , Vol. 162, 111414, 2025. Contribution: Design of a memory-efficient clustering method for big data and real-time streams, based on the combined use of sliding windows, histogram models, and the Wasserstein distance.	
✓	Papers submitted to international conferences	
Jan. 2026	G. Marrakchi, B. Matei. "Phase-Aware Wavelet-Scattering-Based Encoder-Decoder for Dense Predictions". Submitted to The International Conference on Machine Learning (ICML), 2026 . Contribution: Improving dense prediction tasks by preserving phase information in residual connections ("skip connections"). This helps restore the spatial structure, which is often lost in standard scattering transforms.	
Conferences, Summer Schools, & International Collaborations		
June 2026 – July 2026	Invited Researcher Collaboration with: Suela Maxhelaku Building a FHIR interoperability layer for the LCA oncology-AI orchestration framework.	Tirana Albania
July 2025	Conference: Plate-Forme Intelligence Artificielle (PFIA 2025) Covered Topics: "AI & Health" Workshop, Learning Conference (CAp), and National Conference on Artificial Intelligence (CNIA).	Dijon France
July 2024	Summer School: Oxford Machine Learning (OxML) Covered Topics: AI Applied to Health and Biology, Representation Learning, and Generative AI.	Oxford UK
Nov. 2023	Conference: Google France Research Day <i>Participation as an auditor.</i>	Paris France
July 2023	Conference: Plate-Forme Intelligence Artificielle (PFIA 2023) <i>Volunteer on the organizing committee.</i>	Strasbourg France
Volunteering Activities		
Feb. 2024 – Current	Association Française pour l'Intelligence Artificielle (AFIA) – French Organisation of AI Member	France
Oct. 2023 – Current	Association des Masters en Informatique de Galilée (AMIG) – Galilee Master's degrees Organization Member	Paris France
Dec. 2020	ERASMUS Virtual Exchange Initiation to intercultural dialogue facilitation (with Soliya platform).	Online
July 2016 – Dec. 2023	Rotaract Club Sfax El Borj Founding & Advisory Member	Sfax Tunisia
June 2016 – July 2016	Percussionist International musical residency & concert: "Divercities."	Grenoble France
Jan. 2013 – Aug. 2017	Amateur Theater Acting and Direction – 4 plays	
Other Certificates		
Nov. 2018	Université Laval – University of Laval Certificate in Sustainable Development	Online
Mar. 2018	Université Laval – University of Laval Certificate in Responsible Management	Online
Skills		
• Languages: <u>French:</u> Bilingual – <u>English:</u> Fluent (Scientific English / TOEFL B2) – <u>Arabic:</u> Bilingual • Most Relevant Soft Skills: Leadership – Communication • Computer Science (CS) & Artificial Intelligence (AI): ○ <u>Machine Learning:</u> Python – Scikit-learn – TensorFlow – PyTorch ○ <u>Data:</u> SQL (MySQL & Oracle) – PL/SQL – Tableau – SAS – Talend ○ <u>System Administration:</u> C – C++ – Bash – Linux – Raspberry Pi ○ Theoretical Computer Science: Algorithmic – Data Structure		